
Dvoretzky-Milman's Theorem

Here we extend the matrix deviation inequality from Chapter 9 for general norms on $\mathbb{R}^n$, and even for general sub-additive functions on $\mathbb{R}^n$. We use this result to prove the fundamental Dvoretzky-Milman's theorem in high-dimensional geometry. It helps us describe the shape of an m -dimensional random projection of an arbitrary set $T \subset \mathbb{R}^n$. The answer depends on whether m is larger or smaller than the critical dimension, which is the stable dimension $d(T)$. In the high-dimensional regime (where $m \gtrsim d(T)$), the additive Johnson-Lindenstrauss that we studied in Section 9.3.2 shows that the random projection approximately preserves the geometry of T . In the low-dimensional regime (where $m \lesssim d(T)$), geometry can no longer be preserved due to “saturation”. Instead, Dvoretzky-Milman's theorem shows that in this regime the projected set is approximately a *round ball*.

11.1 Deviations of random matrices with respect to general norms

In this section we generalize the matrix deviation inequality from Section 9.1. We replace the Euclidean norm by any positive-homogeneous, subadditive function.

Definition 11.1.1. Let V be a vector space. A function $f : V \rightarrow \mathbb{R}$ is called *positive-homogeneous* if

$$f(\alpha x) = \alpha f(x) \quad \text{for all } \alpha \geq 0 \text{ and } x \in V.$$

The function f is called *subadditive* if

$$f(x + y) \leq f(x) + f(y) \quad \text{for all } x, y \in V.$$

Note that despite being called “positive-homogeneous”, f is allowed to take negative values. (“Positive” here applies to the multiplier α in the definition.)

Example 11.1.2. (a) Any *norm* on a vector space is positive-homogeneous and subadditive. The subadditivity is nothing else than triangle inequality in this case.

(b) Clearly, any *linear functional* on a vector space is positive-homogeneous and subadditive. In particular, for any fixed vector $y \in \mathbb{R}^m$, the function $f(x) = \langle x, y \rangle$ is a positive-homogeneous and subadditive on $\mathbb{R}^m$.

(c) Consider a bounded set $S \subset \mathbb{R}^m$ and define the function

$$f(x) := \sup_{y \in S} \langle x, y \rangle, \quad x \in \mathbb{R}^m. \tag{11.1}$$

Then f is a positive-homogeneous and subadditive on $\mathbb{R}^m$. This function is sometimes called the *support function* of S .

Exercise 11.1.3. ☛ Check that the function $f(x)$ in part c of Example 11.1.2 is positive-homogeneous and subadditive.

Exercise 11.1.4. ☛ Let $f : V \rightarrow \mathbb{R}$ be a subadditive function on a vector space V . Show that

$$f(x) - f(y) \leq f(x - y) \quad \text{for all } x, y \in V. \quad (11.2)$$

We are ready to state the main result of this section.

Theorem 11.1.5 (General matrix deviation inequality). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries. Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be a positive-homogeneous and subadditive function, and let $b \in \mathbb{R}$ be such that*

$$f(x) \leq b\|x\|_2 \quad \text{for all } x \in \mathbb{R}^n. \quad (11.3)$$

Then for any subset $T \subset \mathbb{R}^n$, we have

$$\mathbb{E} \sup_{x \in T} |f(Ax) - \mathbb{E} f(Ax)| \leq Cb\gamma(T).$$

Here $\gamma(T)$ is the Gaussian complexity introduced in Section 7.6.2.

This theorem generalizes the matrix deviation inequality (in the form we gave in Exercise 9.1.2).

Exactly as in Section 9.1, Theorem 11.1.5 would follow from Talagrand's comparison inequality once we show that the random process $X_x := f(Ax) - \mathbb{E} f(Ax)$ has sub-gaussian increments. Let us do this now.

Theorem 11.1.6 (Sub-gaussian increments). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, and let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be a positive homogeneous and subadditive function satisfying (11.3). Then the random process*

$$X_x := f(Ax) - \mathbb{E} f(Ax)$$

has sub-gaussian increments with respect to the Euclidean norm, namely

$$\|X_x - X_y\|_{\psi_2} \leq Cb\|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n. \quad (11.4)$$

Exercise 11.1.7. ☛ Deduce the general matrix deviation inequality (Theorem 11.1.5) from Talagrand's comparison inequality (in the form of Exercise 8.6.4) and Theorem 11.1.6.

Proof of Theorem 11.1.6 Without loss of generality we may assume that $b = 1$. (Why?) Just like in the proof of Theorem 9.1.3, let us first assume that

$$\|x\|_2 = \|y\|_2 = 1.$$

In this case, the inequality in (11.4) we want to prove becomes

$$\|f(Ax) - f(Ay)\|_{\psi_2} \leq C\|x - y\|_2. \quad (11.5)$$

Step 1. Creating independence. Consider the vectors

$$u := \frac{x+y}{2}, \quad v := \frac{x-y}{2} \quad (11.6)$$

Then

$$x = u + v, \quad y = u - v$$

and thus

$$Ax = Au + Av, \quad Ay = Au - Av.$$

(See Figure 11.1).

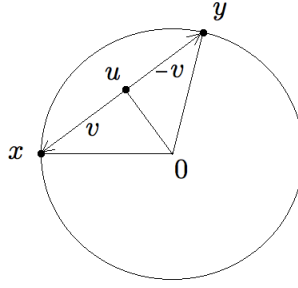


Figure 11.1 Creating a pair of orthogonal vectors u, v out of x, y .

Since the vectors u and v are orthogonal (check!), the Gaussian random vectors Au and Av are independent. (Recall Exercise 3.3.6.)

Step 2. Using Gaussian concentration. Let us condition on $a := Au$ and study the conditional distribution of

$$f(Ax) = f(a + Av).$$

By rotation invariance, $a + Av$ is a Gaussian random vector that we can express as

$$a + Av = a + \|v\|_2 g, \quad \text{where } g \sim N(0, I_m).$$

(Recall Exercise 3.3.3.) We claim that $f(a + \|v\|_2 g)$ as a function of g is Lipschitz with respect to the Euclidean norm on $\mathbb{R}^m$, and

$$\|f\|_{\text{Lip}} \leq \|v\|_2. \quad (11.7)$$

To check this, fix $t, s \in \mathbb{R}^m$ and note that

$$\begin{aligned} f(t) - f(s) &= f(a + \|v\|_2 t) - f(a + \|v\|_2 s) \\ &\leq f(\|v\|_2 t - \|v\|_2 s) \quad (\text{by (11.2)}) \\ &= \|v\|_2 f(t - s) \quad (\text{by positive homogeneity}) \\ &\leq \|v\|_2 \|t - s\|_2 \quad (\text{using (11.3) with } b = 1), \end{aligned}$$

and (11.7) follows.

Concentration in the Gauss space (Theorem 5.2.2) then yields

$$\|f(g) - \mathbb{E} f(g)\|_{\psi_2(a)} \leq C\|v\|_2,$$

or

$$\left\| f(a + Av) - \mathbb{E}_a f(a + Av) \right\|_{\psi_2(a)} \leq C\|v\|_2, \quad (11.8)$$

where the index “ a ” reminds us that these bounds are valid for the conditional distribution, with $a = Au$ fixed.

Step 3. Removing the conditioning. Since random vector $a - Av$ has the same distribution as $a + Av$ (why?), it satisfies the same bound.

$$\left\| f(a - Av) - \mathbb{E}_a f(a - Av) \right\|_{\psi_2(a)} \leq C\|v\|_2. \quad (11.9)$$

Subtract (11.9) from (11.8), use triangle inequality and the fact that the expectations are the same; this gives

$$\left\| f(a + Av) - f(a - Av) \right\|_{\psi_2(a)} \leq 2C\|v\|_2.$$

This bound is for the conditional distribution, and it holds for any fixed realization of a random variable $a = Au$. Therefore, it holds for the original distribution, too:

$$\left\| f(Au + Av) - f(Au - Av) \right\|_{\psi_2} \leq 2C\|v\|_2.$$

(Why?) Passing back to the x, y notation by (11.6), we obtain the desired inequality (11.5).

The proof is complete for the unit vectors x, y ; Exercise 11.1.8 below extends it for the general case. $\square$

Exercise 11.1.8 (Non-unit x, y). $\clubsuit$ Extend the proof above to general (not necessarily unit) vectors x, y . **Hint:** Follow the argument in Section 9.1.4.

Remark 11.1.9. It is an open question if Theorem 11.1.5 holds for general subgaussian matrices A .

Exercise 11.1.10 (Anisotropic distributions). $\clubsuit\clubsuit$ Extend Theorem 11.1.5 to $m \times n$ matrices A whose rows are independent $N(0, \Sigma)$ random vectors, where Σ is a general covariance matrix. Show that

$$\mathbb{E} \sup_{x \in T} \left| f(Ax) - \mathbb{E} f(Ax) \right| \leq Cb\gamma(\Sigma^{1/2}T).$$

Exercise 11.1.11 (Tail bounds). $\clubsuit\clubsuit$ Prove a high-probability version of Theorem 11.1.5. **Hint:** Follow Exercise 9.1.8.

11.2 Johnson-Lindenstrauss embeddings and sharper Chevet inequality

Like the original matrix deviation inequality from Chapter 9, the general Theorem 9.1.1 has many consequences, which we discuss now.

11.2.1 Johnson-Lindenstrauss Lemma for general norms

Using the general matrix deviation inequality similarly to Section 9.3, it should be quite straightforward to do the following exercises:

Exercise 11.2.1. ☛☛ State and prove a version of Johnson-Lindenstrauss Lemma for a general norm (as opposed to the Euclidean norm) on $\mathbb{R}^m$.

Exercise 11.2.2 (Johnson-Lindenstrauss Lemma for ℓ_1 norm). ☛☛ Specialize the previous exercise to the ℓ_1 norm. Thus, let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$, let A be an $m \times n$ Gaussian matrix with i.i.d. $N(0, 1)$ entries, and let $\varepsilon \in (0, 1)$.

Suppose that

$$m \geq C(\varepsilon) \log N.$$

Show that with high probability the matrix $Q := \sqrt{\pi/2} \cdot m^{-1} A$ satisfies

$$(1 - \varepsilon) \|x - y\|_2 \leq \|Qx - Qy\|_1 \leq (1 + \varepsilon) \|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}.$$

This conclusion is very similar to the original Johnson-Lindenstrauss Lemma (Theorem 5.3.1), except the distance between the projected points is measured in the ℓ_1 norm.

Exercise 11.2.3 (Johnson-Lindenstrauss embedding into ℓ_∞). ☛☛ Use the same notation as in the previous exercise, but assume this time that

$$m \geq N^{C(\varepsilon)}.$$

Show that with high probability the matrix $Q := C(\log m)^{-1/2} A$, for some appropriate constant C , satisfies

$$(1 - \varepsilon) \|x - y\|_2 \leq \|Qx - Qy\|_\infty \leq (1 + \varepsilon) \|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}.$$

Note that in this case $m \geq N$, so Q gives an *almost isometric embedding* (rather than a projection) of the set $\mathcal{X}$ into ℓ_∞ .

11.2.2 Two-sided Chevet's inequality

The general matrix deviation inequality will help us sharpen Chevet's inequality, which we originally proved in Section 8.7.

Theorem 11.2.4 (General Chevet's inequality). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries. Let $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$ be arbitrary bounded sets. Then*

$$\mathbb{E} \sup_{x \in T} \left| \sup_{y \in S} \langle Ax, y \rangle - w(S) \|x\|_2 \right| \leq C \gamma(T) \text{rad}(S).$$

Using triangle inequality we can see that Theorem 11.2.4 is a sharper, two-sided form of Chevet's inequality (Theorem 8.7.1).

Proof Let us apply general matrix deviation inequality (Theorem 11.1.5) for the function f defined in (11.1), i.e. for

$$f(x) := \sup_{y \in S} \langle x, y \rangle.$$

To do this, we need to compute b for which (11.3) holds. Fix $x \in \mathbb{R}^m$ and use Cauchy-Schwarz inequality to get

$$f(x) \leq \sup_{y \in S} \|x\|_2 \|y\|_2 = \text{rad}(S) \|x\|_2.$$

Thus (11.3) holds with $b = \text{rad}(S)$.

It remains to compute $\mathbb{E} f(Ax)$ appearing in the conclusion of Theorem 11.1.5. By rotation invariance of Gaussian distribution (see Exercise 3.3.3), the random vector Ax has the same distribution as $g\|x\|_2$ where $g \in N(0, I_m)$. Then

$$\begin{aligned} \mathbb{E} f(Ax) &= \mathbb{E} f(g) \|x\|_2 \quad (\text{by positive homogeneity}) \\ &= \mathbb{E} \sup_{y \in S} \langle g, y \rangle \|x\|_2 \quad (\text{by definition of } f) \\ &= w(S) \|x\|_2 \quad (\text{by definition of the Gaussian width}). \end{aligned}$$

Substituting this into the conclusion of Theorem 11.1.5, we complete the proof. $\square$

11.3 Dvoretzky-Milman's Theorem

Dvoretzky-Milman's Theorem is a remarkable result about random projections of general bounded sets in $\mathbb{R}^n$. If the projection is onto a suitably low dimension, the convex hull of the projected set turns out to be *approximately a round ball* with high probability, see Figures 11.2, 11.3.

11.3.1 Gaussian images of sets

It will be more convenient for us to work with “Gaussian random projections” than with ordinary projections. Here is a very general result that compares the Gaussian projection of a general set to a Euclidean ball.

Theorem 11.3.1 (Random projections of sets). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, and $T \subset \mathbb{R}^n$ be a bounded set. Then the following holds with probability at least 0.99:*

$$r_- B_2^m \subset \text{conv}(AT) \subset r_+ B_2^m$$

where¹

$$r_{\pm} := w(T) \pm C\sqrt{m} \text{rad}(T).$$

¹ As before, $\text{rad}(T)$ denotes the radius of T , which we defined in (8.47).

The left inclusion holds only if r_- is non-negative; the right inclusion, always.

We will shortly deduce this theorem from two-sided Chevet's inequality. The following exercise will provide the link between the two results. It asks you to show that the support function (11.1) of general set S is the ℓ_2 norm if and only if S is the Euclidean ball; there is also a stability version of this equivalence.

Exercise 11.3.2 (Almost Euclidean balls and support functions). ☕☕☕

- (a) Let $V \subset \mathbb{R}^m$ be a closed bounded set. Show that $\text{conv}(V) = B_2^m$ if and only if

$$\sup_{x \in V} \langle x, y \rangle = \|y\|_2 \quad \text{for all } y \in \mathbb{R}^m.$$

- (b) Let $V \subset \mathbb{R}^m$ be a bounded set and $r_-, r_+ \geq 0$. Show that the inclusion

$$r_- B_2^m \subset \text{conv}(V) \subset r_+ B_2^m$$

holds if and only if

$$r_- \|y\|_2 \leq \sup_{x \in V} \langle x, y \rangle \leq r_+ \|y\|_2 \quad \text{for all } y \in \mathbb{R}^m.$$

Proof of Theorem 11.3.1 Let us write the two-sided Chevet's inequality in the following form:

$$\mathbb{E} \sup_{y \in S} \left| \sup_{x \in T} \langle Ax, y \rangle - w(T) \|y\|_2 \right| \leq C \gamma(S) \text{rad}(T).$$

where $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$. (To get this form, use Theorem 11.2.4 for T and S swapped with each other and for A^T instead of A – do this!)

Choose S to be the sphere S^{m-1} and recall that its Gaussian complexity $\gamma(S) \leq \sqrt{m}$. Then, by Markov's inequality, the following holds with probability at least 0.99:

$$\left| \sup_{x \in T} \langle Ax, y \rangle - w(T) \|y\|_2 \right| \leq C \sqrt{m} \text{rad}(T) \quad \text{for every } y \in S^{m-1}.$$

Use triangle inequality and recall the definition of $r_{\pm}$ to get

$$r_- \leq \sup_{x \in T} \langle Ax, y \rangle \leq r_+ \quad \text{for every } y \in S^{m-1}.$$

By homogeneity, this is equivalent to

$$r_- \|y\|_2 \leq \sup_{x \in T} \langle Ax, y \rangle \leq r_+ \|y\|_2 \quad \text{for every } y \in \mathbb{R}^m.$$

(Why?) Finally, note that

$$\sup_{x \in T} \langle Ax, y \rangle = \sup_{x \in AT} \langle x, y \rangle$$

and apply Exercise 11.3.2 for $V = AT$ to complete the proof. $\square$

11.3.2 Dvoretzky-Milman's Theorem

Theorem 11.3.3 (Dvoretzky-Milman's theorem: Gaussian form). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, $T \subset \mathbb{R}^n$ be a bounded set, and let $\varepsilon \in (0, 1)$. Suppose*

$$m \leq c\varepsilon^2 d(T)$$

where $d(T)$ is the stable dimension of T introduced in Section 7.6. Then with probability at least 0.99, we have

$$(1 - \varepsilon)B \subset \text{conv}(AT) \subset (1 + \varepsilon)B$$

where B is a Euclidean ball with radius $w(T)$.

Proof Translating T if necessary, we can assume that T contains the origin. Apply Theorem 11.3.1. All that remains to check is that $r_- \geq (1 - \varepsilon)w(T)$ and $r_+ \leq (1 + \varepsilon)w(T)$, which by definition would follow if

$$C\sqrt{m} \text{rad}(T) \leq \varepsilon w(T). \quad (11.10)$$

To check this inequality, recall that by assumption and Definition 7.6.2 we have

$$m \leq c\varepsilon^2 d(T) \leq \frac{\varepsilon^2 w(T)^2}{\text{diam}(T)^2}$$

provided the absolute constant $c > 0$ is chosen sufficiently small. Next, since T contains the origin, $\text{rad}(T) \leq \text{diam}(T)$. (Why?) This implies (11.10) and completes the proof. $\square$

Remark 11.3.4. As is obvious from the proof, if T contains the origin then the Euclidean ball B can be centered at the origin, too. Otherwise, the center of B can be chosen as x_0 , where $x_0 \in T$ is any fixed point.

Exercise 11.3.5. ☕☕ State and prove a high-probability version of Dvoretzky-Milman's theorem.

Example 11.3.6 (Projections of the cube). Consider the cube

$$T = [-1, 1]^n = B_\infty^n.$$

Recall that

$$w(T) = \sqrt{\frac{2}{\pi}} \cdot n;$$

recall (7.17). Since $\text{diam}(T) = 2\sqrt{n}$, that the stable dimension of the cube is

$$d(T) \asymp \frac{w(T)^2}{\text{diam}(T)^2} \asymp n.$$

Apply Theorem 11.3.3. If $m \leq c\varepsilon^2 n$ then with high probability we have

$$(1 - \varepsilon)B \subset \text{conv}(AT) \subset (1 + \varepsilon)B$$

where B is a Euclidean ball with radius $\sqrt{2/\pi} \cdot n$.

In words, a random Gaussian projection of the cube onto a subspace of dimension $m \asymp n$ is close to a round ball. Figure 11.2 illustrates this remarkable fact.

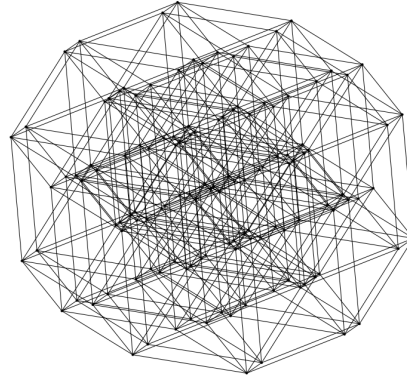


Figure 11.2 A random projection of a 7-dimensional cube onto the plane

Exercise 11.3.7 (Gaussian cloud). ☕ Consider a Gaussian cloud of n points in $\mathbb{R}^m$, which is formed by i.i.d. random vectors $g_1, \dots, g_n \sim N(0, I_m)$. Suppose that

$$n \geq \exp(Cm)$$

with large enough absolute constant C . Show that with high probability, the convex hull of the Gaussian cloud is approximately a Euclidean ball with radius $\sim \sqrt{\log n}$. See Figure 11.3 for illustration.

Hint: Set T to be the canonical basis $\{e_1, \dots, e_n\}$ in $\mathbb{R}^n$, represent the points as $g_i = Te_i$, and apply Theorem 11.3.3.

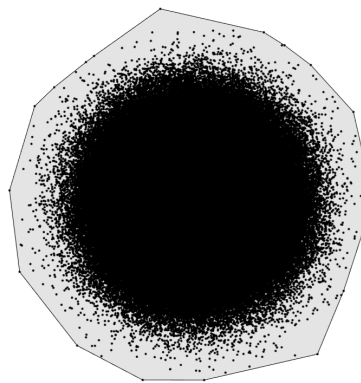


Figure 11.3 A gaussian cloud of 10^7 points on the plane, and its convex hull.

Exercise 11.3.8 (Projections of ellipsoids). 🍷🍷🍷 Consider the ellipsoid $\mathcal{E}$ in $\mathbb{R}^n$ given as a linear image of the unit Euclidean ball, i.e.

$$\mathcal{E} = S(B_2^n)$$

where S is an $n \times n$ matrix. Let A be the $m \times n$ Gaussian matrix with i.i.d. $N(0, 1)$ entries. Suppose that

$$m \lesssim r(S)$$

where $r(S)$ is the stable rank of S (recall Definition 7.6.7). Show that with high probability, the Gaussian projection $A(\mathcal{E})$ of the ellipsoid is almost a round ball with radius $\|S\|_F$:

$$A(\mathcal{E}) \approx \|S\|_F B_2^n.$$

Hint: First replace in Theorem 11.3.3 the Gaussian width $w(T)$ with the quantity $h(T) = (\mathbb{E} \sup_{t \in T} \langle g, t \rangle^2)^{1/2}$, which we discussed in (7.19) and which is easier to compute for ellipsoids.

Exercise 11.3.9 (Random projection in the Grassmannian). 🍷🍷🍷 Prove a version of Dvoretzky-Milman's theorem for the projection P onto a random m -dimensional subspace in $\mathbb{R}^n$. Under the same assumptions, the conclusion should be that

$$(1 - \varepsilon)B \subset \text{conv}(PT) \subset (1 + \varepsilon)B$$

where B is a Euclidean ball with radius $w_s(T)$. (Recall that $w_s(T)$ is the spherical width of T , which we introduced in Section 7.5.2)

Summary of random projections of geometric sets

It is useful to compare Dvoretzky-Milman's theorem to our earlier estimates on the diameter of random projections of geometric sets, which we developed in Sections 7.7 and 9.2.2. We found that a random projection P of a set T onto an m -dimensional subspace in $\mathbb{R}^n$ satisfies a phase transition. In the high-dimensional regime (where $m \gtrsim d(T)$), the projection shrinks the diameter of T by the factor of order $\sqrt{m/n}$, i.e.

$$\text{diam}(PT) \lesssim \sqrt{\frac{m}{n}} \text{diam}(T) \quad \text{if } m \geq d(T).$$

Moreover, the additive Johnson-Lindenstrauss from Section 9.3.2 shows that in this regime, the random projection P approximately preserves the geometry of T (the distances between all points in T shrink roughly by the same scaling factor).

In the low-dimensional regime (where $m \lesssim d(T)$), the size of the projected set surprisingly stops shrinking. All we can say is that

$$\text{diam}(PT) \lesssim w_s(T) \asymp \frac{w(T)}{\sqrt{n}} \quad \text{if } m \leq d(T),$$

see Section 7.7.1.

Dvoretzky-Milman's theorem explains why the size of T stops shrinking for

$m \lesssim d(T)$. Indeed, in this regime the projection PT is *approximately the round ball* of radius of order $w_s(T)$ (see Exercise 11.3.9), regardless how small m is.

Let us summarize our findings. *A random projection of a set T in $\mathbb{R}^n$ onto an m -dimensional subspace approximately preserves the geometry of T if $m \gtrsim d(T)$. For smaller m , the projected set PT becomes approximately a round ball of diameter $\sim w_s(T)$, and its size does not shrink with m .*

11.4 Notes

General matrix deviation inequality (Theorem 11.1.5) and its proof is due to G. Schechtman [181].

The original version Chevet's inequality was proved by S. Chevet [54] and the constant factors there were improved by Y. Gordon [84]; see also [11, Section 9.4], [130, Theorem 3.20] and [205, 2]. The version of Chevet's inequality that we stated in Theorem 11.2.4) can be reconstructed from the work of Y. Gordon [84, 86], see [130, Corollary 3.21].

Dvoretzky-Milman's theorem is a result with a long history in functional analysis. Proving a conjecture of A. Grothendieck, A. Dvoretzky [73, 74] proved that any n -dimensional normed space has an m -dimensional almost Euclidean subspace, where $m = m(n)$ grows to infinity with n . V. Milman gave a probabilistic proof of this theorem and pioneered the study of the best possible dependence $m(n)$. Theorem 11.3.3 is due to V. Milman [148]. The stable dimension $d(T)$ is the critical dimension in Dvoretzky-Milman's theorem, i.e. its conclusion always fails for $m \gg d(T)$ due to a result of V. Milman and G. Schechtman [151], see [11, Theorem 5.3.3]. The tutorial [13] contains a light introduction into Dvoretzky-Milman theorem. For a full exposition of Dvoretzky-Milman's theorem and many of its ramifications, see e.g. [11, Chapter 5 and Section 9.2], [130, Section 9.1] and the references there.

An important question related to Dvoretzky-Milman and central limit theorems is about m -dimensional random projections (marginals) of a given probability distribution in $\mathbb{R}^n$; we may wonder whether such marginals are approximately normal. This question may be important in data science applications, where “wrong” lower-dimensional random projections of data sets in $\mathbb{R}^n$ form a “gaussian cloud”. For log-concave probability distributions, such kind of central limit theorem was first proved by B. Klartag [116]; see the history and more recent results in [11, Section 10.7]. For discrete sets, this see E. Meckes [142] and the references there.

The phenomenon we discussed in the summary in the end of Section 7.7 is due to V. Milman [149]; see [11, Proposition 5.7.1].